

L-18: Distributional Convergence (Asymptotic Distribution)

Example 1: Binomial to Poisson

Let $X_n \sim \text{Bin}(n, \frac{1}{n})$ for $n \geq 1$. We will show that $P(X_n = k) \rightarrow \frac{e^{-1}}{k!}$ as $n \rightarrow \infty$, for all $k \geq 0$. This limiting distribution is the PMF of a Poisson(1) random variable.

Recall the Binomial PMF:

$$P(X_n = k) = \binom{n}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k}$$

Let's break this down:

$$P(X_n = k) = \frac{n(n-1) \cdots (n-k+1)}{k!} \cdot \frac{1}{n^k} \cdot \left(1 - \frac{1}{n}\right)^n \cdot \left(1 - \frac{1}{n}\right)^{-k}$$

We can combine the first two terms by distributing the k factors of n from n^k into the k factors of the numerator:

$$\begin{aligned} \frac{n(n-1) \cdots (n-k+1)}{n^k} &= \left(\frac{n}{n}\right) \left(\frac{n-1}{n}\right) \left(\frac{n-2}{n}\right) \cdots \left(\frac{n-k+1}{n}\right) \\ &= (1) \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \end{aligned}$$

So, we can write the full expression as:

$$P(X_n = k) = \frac{1}{k!} \cdot \left[\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \right] \cdot \left(1 - \frac{1}{n}\right)^n \cdot \left(1 - \frac{1}{n}\right)^{-k}$$

Now, we take the limit as $n \rightarrow \infty$ for each part (for a fixed k):

- $\frac{1}{k!}$ (constant)
- $\left[\left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)\right] \rightarrow (1) \cdots (1) = 1$
- $\left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1}$
- $\left(1 - \frac{1}{n}\right)^{-k} \rightarrow (1)^{-k} = 1$

we know:

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e$$

Combining above things , we get:

$$\lim_{n \rightarrow \infty} P(X_n = k) = \frac{1}{k!} \cdot 1 \cdot e^{-1} \cdot 1 = \frac{e^{-1}}{k!}$$

This shows that X_n "converges" to a Poisson(1) distribution.

This means for large n , the $\text{Bin}(n, 1/n)$ distribution is well-approximated by the Poisson(1) distribution.

Example 2: Discrete Uniform to Continuous Uniform

Let $X_n \sim \text{Discrete Uniform}\{1, 2, \dots, n\}$. Define a new random variable $Y_n = \frac{X_n}{n}$. The range of Y_n is $\{\frac{1}{n}, \frac{2}{n}, \dots, 1\}$.

Euler's geometric formula: $V-E+F=2$

Let's find the CDF of Y_n , $F_n(t) = P(Y_n \leq t)$, for a fixed $t \in [0, 1]$.

$$F_n(t) = P\left(\frac{X_n}{n} \leq t\right) = P(X_n \leq nt)$$

Since X_n is uniform on the integers $\{1, \dots, n\}$, the number of outcomes $\leq nt$ is $\lfloor nt \rfloor$. The total number of outcomes is n .

$$F_n(t) = \frac{\lfloor nt \rfloor}{n}$$

Now, let's find the limit as $n \rightarrow \infty$. We know from the property of the floor function:

$$nt - 1 < \lfloor nt \rfloor \leq nt$$

Dividing by n (for $n > 0$):

$$t - \frac{1}{n} < \frac{\lfloor nt \rfloor}{n} \leq t$$

As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$. By the Sandwich Theorem:

$$\lim_{n \rightarrow \infty} \frac{\lfloor nt \rfloor}{n} = t$$

So, $\lim_{n \rightarrow \infty} F_n(t) = t$.

Let $U \sim \text{Uniform}(0, 1)$. Its CDF is $F_U(t) = P(U \leq t) = t$ for $t \in [0, 1]$. Thus, the CDF of Y_n converges to the CDF of a $\text{Uniform}(0, 1)$ random variable.

Distribution of Y_n "well approximates" the $U(0,1)$ distribution.

Theorem 1 (POWER OF THE LORD Theorem). *Let S be a finite set. Let $X_1, X_2, \dots$ be iid samples from S . Fix $N \geq 1$ and $s_1, s_2, \dots, s_N \in S$.*

$$A_n : [X_n = s_1, X_{n+1} = s_2, \dots, X_{n+N-1} = s_N]$$

$$P(A_n \text{ i.o.}) = ?$$

If we have $U \sim \text{Unif}(0, 1)$, then I can always have $X \sim \text{Unif}(S)$.

$S := \{\lambda_1, \lambda_2, \dots, \lambda_k\}$, where $k = |S|$.

$$X = \begin{cases} \lambda_1 & \text{if } U \leq \frac{1}{k} \\ \lambda_2 & \text{if } \frac{1}{k} < U \leq \frac{2}{k} \\ \vdots & \\ \lambda_k & \text{if } \frac{k-1}{k} < U \leq \frac{k}{k} = 1 \end{cases}$$

So if I have $U_1, U_2, \dots$ iid $\text{Unif}(0, 1)$, this I can get $X_1, X_2, \dots$ iid $\text{Unif}(S)$.

$$P(A_n) = \frac{1}{K^N} > 0$$

$$K = |S|$$

doesn't depend on n

$$\sum_{n=1}^{\infty} P(A_n) = \infty$$

$$P(A_n \text{ i.o.}) = ? = 1$$

if independent then by 2nd BC (Borel-Cantelli) the ans is 1.

$$A_1 = [X_1 = s_1, X_2 = s_2, \dots, X_N = s_N]$$

$$A_2 = [X_2 = s_1, X_3 = s_2, \dots, X_{N+1} = s_N]$$

and so on....

$A_1, A_{N+1}, A_{2N+1}, \dots$ Sum of probabilities of them is also ∞ .

$$P(A_n \text{ i.o.}) = 1$$

when 1st A_n happens?

Geometric $\left(\frac{1}{K^N}\right)$

Expected value? K^N

Approximation of Distribution

(a) $\text{Bin}\left(n, \frac{1}{n}\right) \approx \text{Poisson}(1)$

(b) $D \text{ U}\{1, 2, \dots, n\} \approx \text{Uniform}(0, 1)$

seen one more!

fix a p in $(0, 1)$

$\text{Bin}(n, p) \approx \text{Normal}(np, np(1-p))$

Definition! Let $\{F_n\}$ be a sequence of PDFs and F be another PDF. We say F_n converges to F

$$F_n \Rightarrow F$$

If $\forall x \in \mathbb{R}$ s.t. x is a point of continuity of F .

$$F_n(x) \longrightarrow F(x) \quad \text{as } n \rightarrow \infty$$

Often when we take r.v.s (random variables), say $X_n \sim F_N$ r.v.s and $X \sim F$ r.v., we will write

$$X_n \xrightarrow{d} X$$

and say, " X_n converges in distribution to X "

$$F_n(x) = P(X_n \leq x), x \in \mathbb{R}$$

$$F(x) = P(X \leq x)$$

$$X_n \xrightarrow{d} X$$

where $X \sim \text{Pois}(1)$.

pdf of X ?

e^{-1} for $X=1$

$e^{-1} + e^{-2}$ for $X=2$

$P(X_n \leq t)$ t is not an integer.

$$= \sum_{k=0}^{\lfloor t \rfloor} P(X_n = k) \longrightarrow \sum_{k=0}^{\lfloor t \rfloor} P(X = k) = P(X \leq t)$$

$\forall B \in \mathbb{R}$

$$P\left(\frac{X_n - np}{\sqrt{np(1-p)}} \leq b\right) \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^b e^{-t^2/2} dt = \Phi(b)$$

$$\frac{X_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} Z \quad \text{where } Z \sim N(0, 1)$$

De Moivre-Laplace Central Limit Theorem